

Web Technology Fundamentals: Comprehensive Study Guide

Unit 1: HTML and CSS

HTML Basics

HTML Tag Reference: Understanding the fundamental building blocks of web pages using tags.

Global Attributes: Attributes applicable to any HTML element (e.g., id, class, style, title).

Document Structure Tags: Essential tags defining the overall structure of an HTML document (<html>, <head>, <title>, <body>).

Formatting Tags: Tags used for basic text formatting (, , <i>, , <u>,
, <hr>).

Text-Level and Block-Level Formatting: Differentiating between inline (, <abbr>, <cite>, <code>) and block-level (<div>, <p>, <h1> to <h6>) elements.

Lists: Creating ordered (,), unordered (,), and definition lists (<dl>, <dt>, <dd>).

Hyperlinks: Using the anchor tag (<a>) to create links.

Executable Content Tags: Embedding scripts (<script>) and providing fallback content (<noscript>).

Image and Imagemaps

Inserting Images: Embedding images using the tag with attributes like src, alt, width, height.

Client-Side Image Maps: Creating clickable regions on an image using <map> and <area>.

Server-Side Image Maps: Using the ismap attribute for server-processed image maps.

Tables

Basic Tags: Structure of tables (<table>, <tr>, <td>, <th>) and captions (<caption>).

Attributes: Common table attributes (e.g., border, cellpadding, cellspacing, bgcolor, width, align).

Merging Cells: Using rowspan and colspan for complex table layouts.

Table Sections: Grouping table content with <thead>, <tbody>, <tfoot>.

Tables for Design: Historical use of tables for page layout.

Frames

Tags: Defining frame layouts (<frameset>, <frame>) and inline frames (<iframe>).

Usage: Targeting frames, nesting framesets, and hidden frames.

Forms

Creating Forms: The <form> tag with action and method attributes.

Input Types: Various input field types (e.g., text, password, email, radio, checkbox, submit, file, textarea).

Grouping & Labeling: Enhancing form structure and accessibility with <fieldset>, <legend>, and <label>.

Attributes: Common input attributes (e.g., disabled, readonly, required, maxlength, placeholder).

Form Event Handlers: Basic event handlers (onsubmit, onchange, onfocus).

Style Sheets (CSS)

Purpose: Controlling the presentation and layout of HTML elements.

Types of CSS: Inline, Internal, and External CSS.

Cascading Order: Understanding the precedence rules for applying styles.

Unit 2: Issues of Web Technology

Architectural Issues

Web architecture: Components and interactions within a web application (client, server, database).

Tier Technology: 2-Tier: Direct client-server/database communication.

3-Tier: Client, Application Server, Database Server.

n-Tier: Multiple layers for scalability.

Unit 3: The Client Tier

XML (Extensible Markup Language)

Basic Rules: Well-formed XML requirements (root element, closed tags, case-sensitivity, quoted attributes).

Namespaces: Avoiding naming conflicts.

XML Schema (XSD)

Purpose: Describing XML document structure and data types.

Types and Facets: Simple and complex types, constraints (e.g., minLength, maxLength, pattern).

Occurrence and Order Indicators: Controlling element repetition and sequence (minOccurs, maxOccurs, all, choice, sequence).

DTD (Document Type Definition)

Types: Internal, External, and Public DTDs.

XSL/XSLT

Purpose: Styling and transforming XML documents.

XPath, XQuery

XPath: Navigating elements and attributes within an XML document.

XQuery: Querying XML data.

DOM vs SAX

DOM (Document Object Model): In-memory representation of an XML document.

SAX (Simple API for XML): Event-based parsing, efficient for large files.

Creating XMI

XMI (XML Metadata Interchange): Used for modeling, less common in direct web development.

Unit 4: The Server Tier

Web Server Concept

Role: Hosting web applications (e.g., Apache, Nginx).

Dynamic Content

Generation: Content generated on request using server-side languages.

Control Flow

Logic: Use of if-else, loops for dynamic output.

Session and State

Session: Storing temporary user data on the server.

Cookies: Storing small data on the client.

Error Handling

Management: Using try-catch blocks or custom error pages.

Tag Libraries

Reusable Components: Custom tags for server-side templating (e.g., JSP Tag Libraries).

Unit 5: Advanced Server-Side Issues

Database Connectivity

Methods: Connecting to databases using server-side languages.

SQL Statements

CRUD Operations: SELECT, INSERT, UPDATE, DELETE commands.

Authentication

Types: Anonymous, IP-based, Windows Authentication.

Cookies

Usage: Storing and retrieving small client-side data.

File Handling

Operations: Opening, reading, and writing files on the server.

Form Handling

Processing: Retrieving data via POST/GET, validation, and processing.

Quiz: Web Technology Fundamentals

Instructions: Answer each question in 2-3 sentences.

What is the primary purpose of HTML, and how does it define elements?

Explain the difference between a text-level formatting tag and a block-level formatting tag in HTML, providing an example of each.

Describe the three main types of CSS (Inline, Internal, External) and briefly explain their placement within an HTML document.

What is the purpose of the `<form>` tag in HTML, and which two essential attributes are typically used with it?

Briefly explain the concept of client-side image maps. What two HTML tags are crucial for implementing them?

Distinguish between a 2-Tier and a 3-Tier web architecture.

List two fundamental rules that XML documents must adhere to for them to be considered "well-formed."

What is the main function of XML Schema (XSD), and how does it differ from a DTD in terms of describing XML structure?

Explain the difference between DOM and SAX parsers when processing XML documents, particularly concerning memory usage.

Describe the concept of "dynamic content" in the context of the server tier, and mention one way it is generated.

Quiz Answer Key

HTML (HyperText Markup Language) is the standard language for creating web pages. It defines elements using tags enclosed in angle brackets, such as `<html>` or `<body>`, to structure the content.

Text-level formatting tags (`<span>`, `<em>`) apply styling or meaning to a small portion of text without creating new lines, while block-level formatting tags (`<div>`, `<p>`) create a new block that typically starts on a new line and takes up the full width available.

Inline CSS is applied directly within an HTML tag using the `style` attribute. Internal CSS is defined within `<style>` tags in the `<head>` section of an HTML document. External CSS is linked from an external `.css` file using the `<link>` tag in the `<head>`.

The `<form>` tag is used to create an HTML form for user input. Its essential attributes are `action`, which specifies where to send the form data upon submission, and `method`, which defines the HTTP method (e.g., `post`, `get`) for sending the data.

Client-side image maps allow specific regions of an image to be clickable hyperlinks. They are implemented using the `<map>` tag to define the map and its `<area>` tags for the clickable regions, and the `<img>` tag uses the `usemap` attribute to link to the map.

In a 2-Tier architecture, the client communicates directly with the server or database. In contrast, a 3-Tier architecture introduces an application server between the client and the database server, providing an additional layer for logic and processing.

Two fundamental rules for well-formed XML documents are: they must have a single root element that encloses all other elements, and all tags must be properly closed, meaning every opening tag must have a corresponding closing tag.

XML Schema (XSD) describes the structure and data types of an XML document, offering more robust data typing and validation capabilities compared to DTD. While DTDs define the elements and attributes, XSDs can specify data types, ranges, and patterns for content.

DOM (Document Object Model) parsers load the entire XML document into memory, creating a tree-like structure, which is good for navigation and manipulation but can be memory-intensive for large files. SAX (Simple API for XML) parsers are event-based, processing the document sequentially without loading it entirely, making them more efficient for very large XML files.

Dynamic content refers to web page content that is generated on the fly in response to a user's request, rather than being a static pre-written file. This content is typically generated using server-side languages like PHP or Python, which process data from databases or user input to create tailored output.

Essay Format Questions

Compare and contrast the different types of CSS (Inline, Internal, External) in terms of their implementation, advantages, and disadvantages. Discuss when each type would be most appropriately used in web development.

Discuss the evolution of web page layout from using HTML tables for design to the modern reliance on CSS. Explain the problems associated with using tables for layout and how CSS addresses these issues, improving maintainability and accessibility.

Explain the importance of forms in web applications. Describe the various input types available in HTML forms and discuss how server-side processing is essential for handling and validating the data submitted through these forms.

Analyse the role of XML in web technology. Discuss its basic rules, and then differentiate between DTD and XML Schema (XSD) in terms of their purpose and capabilities for defining XML document structure.

Elaborate on the different architectural tiers (2-Tier, 3-Tier, n-Tier) in web applications. For each tier, describe its components, the flow of data, and the primary benefits it offers in terms of scalability, security, and maintainability.

Glossary of Key Terms

Anchor Tag (<a>): An HTML tag used to create hyperlinks, connecting one web page or resource to another.

Attribute: A modifier of an HTML element, providing additional information about the element (e.g., src for , href for <a>).

Block-Level Element: An HTML element that creates a new block of content, typically starting on a new line and taking up the full available width (e.g., `<div>`, `<p>`, `<h1>`).

Cascading Style Sheets (CSS): A styling language used to describe the presentation of HTML documents, controlling layout, colours, fonts, and more.

Client-Side Image Map: A clickable image where the regions and their associated links are defined within the HTML on the client's browser.

Cookies: Small pieces of data stored on the user's web browser by a website, often used to remember stateful information or user preferences.

Database Connectivity: The process of establishing a connection between a server-side application and a database to store and retrieve data.

Definition List (`<dl>`): An HTML list type used to define terms (`<dt>`) and their descriptions (`<dd>`).

Document Object Model (DOM): A programming interface for HTML and XML documents, representing the page as a tree structure of objects, allowing programs to change document structure, style, and content.

Document Type Definition (DTD): A set of markup declarations that define a valid document type for an SGML-family markup language (like XML or HTML).

Dynamic Content: Web page content that is generated on the server at the time of the user's request, often based on user input or database queries.

External CSS: A method of applying CSS where styles are defined in a separate .css file and linked to the HTML document.

Form: An HTML element (`<form>`) used to collect user input, typically containing various input fields, buttons, and other controls.

Frame: An HTML feature (now largely deprecated for `<iframe>`) that allowed a web page to be divided into multiple independent sections, each displaying a separate HTML document.

Global Attributes: HTML attributes that can be applied to any HTML element (e.g., `id`, `class`, `style`, `title`).

HyperText Markup Language (HTML): The standard markup language for creating web pages and web applications.

Inline CSS: A method of applying CSS where styles are defined directly within an HTML element's style attribute.

Inline Frame (`<iframe>`): An HTML tag used to embed another HTML document within the current HTML document.

Input Type: An attribute of the `<input>` tag that specifies the type of data to be collected (e.g., `text`, `password`, `radio`, `submit`).

Internal CSS: A method of applying CSS where styles are defined within a `<style>` tag in the `<head>` section of an HTML document.

Ordered List (): An HTML list where items are presented in a numbered or lettered sequence.

SAX (Simple API for XML): An event-based API for parsing XML documents, which processes the document sequentially without loading the entire document into memory.

Server-Side Scripting: Programming languages (e.g., PHP, Python, Java) executed on a web server to generate dynamic web content, interact with databases, and handle form submissions.

Session: A mechanism for storing temporary user-specific data on the server during a user's interaction with a web application.

SQL (Structured Query Language): A domain-specific language used in programming and designed for managing data held in a relational database management system.

Tag: A keyword in angle brackets (e.g., <body>, <p>) that defines an element in HTML or XML.

Text-Level Element: An HTML element that applies styling or meaning to a small portion of text without creating new lines (e.g., , , <code>).

Tier Technology: Refers to the architectural layers in a web application, such as 2-tier (client-server/database) or 3-tier (client-application server-database server).

Unordered List (): An HTML list where items are presented with bullet points or other markers.

Web Server: A computer program that stores web content (like HTML pages, images, and videos) and delivers it to web browsers upon request.

XML (Extensible Markup Language): A markup language designed to store and transport data, allowing developers to define their own tags.

XML Schema Definition (XSD): A W3C recommendation that specifies how to formally describe the elements and attributes that can appear in an XML document, along with their data types and relationships.

XPath: A query language for selecting nodes or node-sets in an XML document.

XSLT (Extensible Stylesheet Language Transformations): A language for transforming XML documents into other XML documents, or into other formats such as HTML for web pages, or plain text.